WebClient Computing

Document Object Model (DOM)

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Is DOM Provided by HTML or Javascript

- A question from the last class
- DOM is not part of Javascript

The **Document Object Model** (*DOM*) is a programming interface for HTML,

Although the DOM is often accessed using JavaScript, it is not a part of the JavaScript language. It can also be accessed by other languages.

Excerpt from https://developer.mozilla.org/en-US/docs/Web/API/Document_Object_Model

Accessing Element

- document.getElementById(“ID_NAME”)
  - Returns an element with a given ID_NAME
  - Duplicate ID is not allowed in the HTML specification
- dom_element_by_id.html

```
> var t = document.getElementById("first-p");
<· undefined
> t
<· ▼<p id="first-p">
      "The "
      <strong>first</strong>
      " paragraph"
    </p>
> t.innerHTML
<· "The <strong>first</strong> paragraph"
> t.innerText
<· "The first paragraph"
```

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM test</title>
</head>
<body>
  <p id = "first-p">The <strong>first</strong> paragraph</p>
  <p id = "second-p">The second paragraph</p>
  <p id = "first-p">Duplicated ID paragraph</p>
</body>
</html>
```

Changing Values of an Element

- Assign a value to an element
- Change style of an element
  - Using element.style property

```
> t.innerText = "Changing content";
<· "Changing content"
> t.style.fontSize = "3em";
<· "3em"
> t.style.color="red";
<· "red"
```

Changing content

The second paragraph

Duplicated ID paragraph

Access Elements Other Than ID

- document.getElementsByClassName
  - Get a list of elements based on the class name
- document.getElementsByName
  - Get a list of elements based on the name
  - Name is generally used with input tag
- document.getElementsByTagName
  - Get a list of elements with the input tag name

Accessing Elements Examples

```
> t=document.getElementsByClassName("class-1");
<· [  <p class="class-1">Class 1 paragraph</p>,
    <div class="class-1">Class 1 div</div>]
> t.length
<· 2
> t[0]
<·   <p class="class-1">Class 1 paragraph</p>
> t[1]
<·   <div class="class-1">Class 1 div</div>
```

```
> t=document.getElementsByTagName("p");
<· [  <p class="class-1">Class 1 paragraph</p>,
    <p class="class-2">Class 2 paragraph</p>]
```

```
> t=document.getElementsByName("text-input");
<· [  <input type="text" name="text-input">,
    <input type="text" name="text-input">]
> t[0]
<·   <input type="text" name="text-input">
> t[1].value = "Kyungyong";
<· "Kyungyong"
> t[1].value;
<· "Lee"
```

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM test</title>
</head>
<body>
  <p class = "class-1">Class 1 paragraph</p>
  <p class = "class-2">Class 2 paragraph</p>
  <div class = "class-1">Class 1 div</div>
  ID: <input type="text" name="text-input">
  Name: <input type="text" name="text-input">
</body>
</html>
```

dom_elements_by.html

Adding New Element

- createElement(“element name”);
  - Create a new element to be added in DOM
- createTextNode(“text string for a value”);
  - Create a text node that contains text information
- appendChild(NODE)
  - Append the input node as a child node

Adding New Element Example

```
> var p_elem = document.createElement("p");
<· undefined
> var content = document.createTextNode("new strings");
<· undefined
> p_elem
<·   <p></p>
> content
<·   "new strings"
> document.body.appendChild(p_elem);
<·   <p></p>
> p_elem.appendChild(content);
<·   "new strings"
```

Removing Elements

- removeChild(NODE)
  - Remove the input element from the document
- add_new_element.js and dom_element_by_id.html

```
var p_elem = document.createElement("p");
var content = document.createTextNode("new strings");
p_elem.appendChild(content);

var target = document.getElementById("first-p");

function add() {
  target.appendChild(p_elem);
}

function rm() {
  target.removeChild(p_elem)
}

var add_id = setInterval(add, 3000);
var rm_id = setInterval(rm, 7000);
```

```
> document.body.removeChild(p_elem);
<·   <p>new strings</p>
```

DOM Events

- A Javascript function can be registered to be executed when an event occurs
- Example of HTML events
  - A user clicks a mouse
  - When a image has been loaded
  - When the mouse moves over an element
  - When an input field is changed
  - When a HTML form is submitted

DOM Event Example

- event.html and event.js

```html
<!DOCTYPE html>
<html>
<head>
  <title>Javascript event</title>
  <script type="text/javascript" src="event.js"></script>
</head>
<body>

<p>Click the button to display the date.</p>

<button onclick="displayDate()">The time is?</button>

<p id="demo"></p>

</body>
</html>
```

```javascript
function displayDate() {
    document.getElementById("demo").innerHTML = Date();
}
```

Example of DOM Events

- onclick – when a mouse is clicked
- ondblclick – when a user double clicks a mouse
- onmouseenter – when a mouse pointer enters area
- onmouseleave – when a mouse pointer leaves area
- onkeydown – when a user presses a key
- onload – fires when a page is completely loaded
- Other events - http://www.w3schools.com/jsref/dom_obj_event.asp

DOM Events Example

- onclick and ondblclick
- event_mouse.js and event_mouse.html

```html
<!DOCTYPE html>
<html>
<head>
  <title>Javascript event</title>
  <script type="text/javascript" src="event_mouse.js"></script>
</head>
<body>

<p>Click the button to display the date.</p>

<button onclick="displayDate()">The time is?</button>

<p id="demo"></p>

<p id="dbl-click" ondblclick="doubleClicked()">Please double click me!!</p>
</body>
</html>
```

References

- Javascript, the Definitive Guide
- w3schools.com
- tryhelloworld.com